

PLANRS ACADEMY

GRADE 1 SCIENCE LESSON GUIDE

COURSE TITLE

Sink or Float Predict & Test

TARGET GRADE

Grade 1 (Age 6-7 Years)

Prita Sciencer (Foundational Stage)

SESSION CONFIGURATION

- **Session Number:** Session 1 (30-Minute Core)
- **Document Code:** PA-SCI-G1-S01-LG
- **Scope:** Complete 10-15 Page Comprehensive Lesson Guide & Educator Script

1. Pedagogical Framework & Objectives

This document serves as the master instructional lesson guide for Planrs Academy's Grade 1 Science program. Designed specifically around active learning, inquiry-based pedagogical frameworks, and the Indian National Curriculum Framework (NCF) for the Foundational/Early Primary stage, this lesson aims to cultivate experimental curiosity, systematic observation, and primitive hypothesis formulation among young learners.

1.1 Core Learning Objectives

- **Cognitive (Knowledge):** Define the states of 'sinking' (Doobna / डूबना) and 'floating' (Tairna / तैरना) using descriptive, child-friendly parameters rather than formal density equations. Accurately categorize 8 typical household items into their corresponding interactive categories based on empirical observation.
- **Psychomotor / Skill-Based:** Formulate individual, explicit predictions regarding an item's state prior to immersion. Execute experiential, water-based tests safely under minimal adult guidance. Systematically record data outcomes using direct check/cross visual mapping tools.
- **Affective / Behavioral:** Cultivate resilience and scientific adaptation when initial intuitive guesses diverge from absolute physical experimental outcomes. Encourages asking exploratory "Why" questions to establish an early analytical mindset.

CRITICAL PEDAGOGICAL CONSTRAINT

Do NOT under any circumstances introduce formal technical jargon such as "Density", "Buoyancy", "Volume", or "Archimedes' Principle". For 6-7 year-old cognitive cohorts, these concepts must be translated completely into sensory-accessible frameworks like "heavy for its size" versus "light for its size".

1.2 Session Parameters

Parameter	Specification	Target Metrics
Total Duration	30 Minutes Active Instruction	100% Core Competency Coverage
Difficulty Setting	Beginner / Foundational	Zero Pre-requisite Knowledge required
Delivery Methodology	Interactive Story-Led + Hands-On Activity	>85% Class Engagement/Retention
Language Tier	Simple, Active Indian English / Hinglish	Max 8-10 words per phrase block

2. Materials, Apparatus & Setup Architecture

To maximize engagement while minimizing friction across diverse tier-1, tier-2, and tier-3 configurations within the Indian demographic, the required materials are split into common home-accessible assets.

2.1 Laboratory Apparatus Requirements

Every student (or instructional presenter) must assemble the "Planrs Magic Science Box" prior to starting the module. The box contains the following explicit materials:

- **The Water Container:** A wide, transparent glass or clear plastic bowl. Must be wide enough to support multiple items simultaneously without overcrowding (minimum diameter recommendation: 15-20cm).
- **The Baseline Towel:** A dry cotton towel or kitchen cloth (Unchaa / Rumaal) positioned underneath the basin to immediately handle incidental splashes or water drips during the "Testing Time" sequence.
- **The Learner's Logbook:** The printed Planrs Lab Book or a simple ruled notebook with a pencil for capturing prediction markers (Smileys, ticks, or rudimentary illustrations).

2.2 Targeted Test Objects Strategy

Objects are precisely categorized based on their structural property outcomes to validate both expected archetypes and cognitive conflicts:

Object Code	Object Name (Indian Context)	Material Property	Expected Outcome
OBJ-01	Metal Spoon (Steel Chammach)	Solid Heavy Stainless Steel	SINK
OBJ-02	Dried Marigold Flower (Genda Phool)	Organic Light Petals	FLOAT
OBJ-03	Steel Katori (Empty Bowl)	Hollow Metallic Shape	FLOAT (Paradox Anchor)
OBJ-04	Glass Marble (Kancha)	Dense Solid Compact Glass	SINK
OBJ-05	Plastic Bottle Cap	Light Synthetic Polymers	FLOAT
OBJ-06	Fresh Potato (Aaloo)	Dense Solid Tuber Organic	SINK
OBJ-07	Dry Coconut Shell Pieces	Fibrous Curved Structure	FLOAT
OBJ-08	5-Rupee Coin	Solid Nickel-Brass Alloy	SINK

SAFETY PRECAUTION PROTOCOL

Ensure that no sharp objects (such as iron needles or glass shards) are recommended. All metals must be rounded (like a standard puja bell or coin). Parents must be present to prevent young learners from accidentally drinking the testing water or putting small items like marbles in their mouths.

3. Comprehensive Session Timeline

The lesson structure is divided into five segments, optimized around the limited attention span of Grade 1 learners. Each segment has dedicated delivery vectors and precise checkpoints.

3.1 Master Time Allocation Blueprint

- **00:00-05:00 | Segment 1: The Narrative Hook & Context Set**

Focus: Emotional engagement and real-world linkage. The mascot introduces a sudden dilemma in a local setting to spur curiosity.

- **05:00-10:00 | Segment 2: Definition Frameworks & Active Concept Calibration**

Focus: Semantic structuring. Visual definitions of the core terminology paired with immediate on-screen micro-quizzes to check comprehension.

- **10:00-22:00 | Segment 3: The Experiential Laboratory (Hands-On Cycle)**
Focus: Active inquiry. Students cycle through physical objects, making individual predictions before testing them in real-time.
- **22:00-27:00 | Segment 4: Gamified Practice & Interactive Sort Classification**
Focus: Synthesis and verification. Digital drag-and-drop mechanics to lock in categorization rules across abstract concepts.
- **27:00-30:00 | Segment 5: Metacognitive Summary, Badging Ceremony & Homework**
Focus: Retention and closing. Awarding achievements and setting up parent-supported home challenges.

3.2 Interactive Checkpoint Matrix

Timestamp	Type of Checkpoint	Trigger Event / Question	Success Benchmark
08:15	Mid-Concept MCQ	"If a leaf stays on top of the river, is it sinking or floating?"	>90% Accuracy
13:40	Prediction Input	App pauses for 3 seconds: Input prediction for Stainless Steel Spoon.	100% User Selection
24:30	Sorting Game	Drag 6 randomized household objects into proper destination barrels.	>80% Score First-Try

4. Educator Script & Media Production Directives (Mins 0-10)

This script dictates exact speaker dialogue, facial cues, stage actions, and background animation synchronization markers.

4.1 Segment 1: The Narrative Hook & Story Integration (00:00-05:00)

SCENE START

A bright, warm, animated background representing a traditional Indian open courtyard (Aangan). Chiku the Monkey mascot is sitting near a giant blue plastic bucket filled with water. The camera pans down to meet the teacher standing next to Chiku.

***Teacher:** "Namaste my brilliant young scientists! Welcome to Planrs Academy! Today, my super-friendly helper Chiku and I were playing right here in the sunny courtyard. But look what happened! Chiku was playing with his favorite plastic cricket ball, and oops! It slipped right out of his hands and went... *SPLASH* into our bucket!"*

Teacher: *[Gasp dramatically, leaning closer to the camera] "But wait! Look closely! The big plastic ball didn't go to the bottom. It is sitting right on top, swimming happily! Chiku then tried to throw a shiny 5-rupee coin to see if it would swim too. And... *CHHAPPAK!* It dropped straight down to the very dark bottom!"*

Teacher: *"Chiku is scratching his head! Why did the big ball stay on top, while the tiny coin went all the way down? Do you want to help Chiku solve this grand mystery? Let's begin our scientific mission today!"*

4.2 Segment 2: Definition Frameworks & Concept Building (05:00-10:00)

ANIMATION CUE

The camera transitions cleanly to a split-screen graphic template. The left side is labeled "FLOAT" with light blue water waves; the right side is labeled "SINK" with a sandy, rocky floor texture.

Teacher: *"Before we test our toys, let's learn our secret scientific words. Look at the word on the screen: FLOAT (Tairna). Can you say it with me? Float! Floating means when we place something in water, it stays right on top or near the surface. Just like a beautiful, light marigold flower petal from a puja garland!"*

Teacher: *"Now look at the other word: SINK (Doobna). Let's say it out loud together: Sink! Sinking means when we place an object in water, it drops all the way down to the very bottom. Like a heavy stone from the garden path."*

PRODUCER AUDIO DIRECTIVE

Inject distinct, high-impact SFX here. Sinking must trigger a wet "plop/thud" sound, and floating must trigger an airy, musical "ding" sound. This multi-sensory approach keeps younger cohorts engaged.

5. The Hands-On Experiential Laboratory Script (Mins 10-22)

MEDIA DIRECTIVE

The teacher profile minimizes to a corner bubble. The main screen shows a real-time, overhead view of a transparent experimental bowl filled with clean water, resting over a protective cloth towel.

Teacher: "Now, my helpers, it is time to turn your own desks into a real live laboratory! Look inside your Planrs Magic Science Box. Do you see your clear water bowl? Make sure an adult has helped you fill it up halfway. Is your towel underneath to catch spills? Perfect!"

Teacher: "Let's take out our very first object: The heavy, shiny metal spoon—our Steel Chammach. Hold it in your hands. Feel how cold and solid it is. Now, stop! Don't drop it yet! A true scientist always makes a Prediction first. A prediction is your smart guess!"

INTERACTIVE APPLICATION PROMPT

The interactive system pauses video playback entirely. A full-screen UI overlay appears with two oversized tactile choice buttons: [SPOON WILL FLOAT] or [SPOON WILL SINK]. A 3-second musical countdown timer plays.

Teacher: [Resuming once the student registers an option] "Great guessing! Now, place your spoon gently on top of the water surface... and let go! Wow! Look at that! It went straight down like an anchor! The spoon sinks! Did your prediction match? If yes, give yourself a high-five! If not, don't worry—that is how we learn!"

Teacher: "Let's try Object Number 2. Pick up your dry green leaf or your marigold flower petal. It feels super light and soft, doesn't it? Tap your screen right now: Will it sink down to meet the spoon, or float on top with the bubbles?"

ANIMATION CUE

Interactive choice buttons flash on screen for Leaf properties. Once selected, the clip shows the leaf being placed flat onto the water surface.

Teacher: "Drop it in! Look! It is resting on the surface like a little green boat. The leaf floats! The water is holding it up easily because it is light and spread out!"

6. Advanced Cognitive Conflict: The Katori Paradox

This segment introduces a deliberate cognitive conflict designed to break basic weight misconceptions and transition students into deeper visual observation skills.

SCENE CUE

The teacher returns to the center frame, holding up an empty stainless steel Katori bowl in one hand, and a tiny, dense glass marble (Kancha) in the other.

Teacher: "Now, my scientists, it's time for our super challenge question! I have a tiny glass marble and a much larger, heavier steel Katori. If we think heavy things always sink and light things always float, what will happen here? Let's predict! Will this big, heavy metal Katori sink down to the bottom?"

CONCEPTUAL CONFLICT PAUSE

The screen presents choices for the Katori bowl. Most students will instinctively choose "SINK" because the object is made of metal and is visibly much heavier than the leaf or plastic cap.

Teacher: "Let's test it! Watch carefully. I am placing the empty steel Katori gently flat on top of the water... Look! It's floating! Wow! It floats beautifully just like a boat, even though it is made of metal! Why do you think that happened?"

Teacher: [Tilt the Katori slightly on screen so water rushes over the rim] "But look! If I tilt it and let the water rush inside, or if I fill it up with our heavy glass marbles... oh no! Down it goes! *CHHAPPAK!* It sinks straight to the bottom!"

Teacher: "This teaches us a secret rule: It's not just about how heavy an object is! The shape matters too! When the Katori is empty and shaped wide like a bowl, it traps air inside, which helps it float on top!"

PEDAGOGICAL INSIGHT FOR DEVELOPERS

This section is vital for building critical thinking. The visual shift from an empty floating bowl to a filled sinking bowl allows children to grasp structural displacement concepts visually without needing complex mathematical definitions.

7. Gamified Classification Modules (Mins 22-27)

Following the hands-on session, the learner transitions directly into digital application design intended to test categorization retention under variable criteria.

7.1 "Help Chiku Clean the Courtyard" Interface Specs

This module features an interactive drag-and-drop game. The screen depicts two large target barrels located on opposite sides of Chiku's courtyard:

- **Left Barrel: Labeled "FLOAT"** — Decorated with wavy blue water animations, floating bubbles, and a smiling cloud icon.
- **Right Barrel: Labeled "SINK"** — Decorated with dark sand layers, sunken anchor icons, and underwater stones.

7.2 Sorting Queue Assets & Logic Responses

Six item assets appear sequentially at the center bottom of the UI deck. The application handles student actions based on this logic rule table:

Item Asset	Correct Target	Success Action (Correct Drop)	Error Remediation Action (Incorrect Drop)
Rubber Eraser	SINK Barrel	Plays happy chime; item drops cleanly to the bottom.	Plays bouncy "boing" sound; snaps back. Chiku says: "Oops! Touch it, it's solid and heavy for its size! Try again!"
Wooden Ice-Cream Stick	FLOAT Barrel	Plays sound effect; stick bobs up and down on waves.	Item bounces out. Prompt: "Wood loves to swim! Put it on top of the water waves!"
Metallic Key	SINK Barrel	Plays clinking coin sound effect; asset falls flat instantly.	Snaps back. Voice-over: "Keys are made of heavy metal! Where do heavy solid things instantly go?"
Natural Wine Cork	FLOAT Barrel	Plays whistle sound effect; cork floats horizontally.	Snaps back to start. Visual helper points an arrow directly to the Float wave barrel.

DEVELOPMENTAL INTERACTIVE ASSIGNMENT

To account for limited motor precision in Grade 1 users, the drop target boxes must be expansive, covering at least 30% of total screen real estate per barrel. Embellish objects with bright, cartoon-style thick outlines to make them pop against background assets.

8. Summary, Rewards & Outro Directives (Mins 27-30)

The closing sequence focuses on reinforcement, awarding accomplishments, and extending learning into the child's home environment.

8.1 Final Conceptual Summary

The teacher reappears on screen to tie together the core takeaways into a clear framework:

- Objects that are **light for their size**, or shaped wide to trap air inside, will **FLOAT** on top of the water surface.
- Objects that are **solid and heavy for their size** will **SINK** all the way down to the bottom of the basin.
- Real scientists always **look, predict, and test** to find out how the world works!

8.2 The Junior Water Scientist Badging Ceremony

BADGING EVENT TIMELINE

As the sorting game finishes, the screen darkens slightly, and a colorful animated badge emerges from a spinning star field. The text reads: **JUNIOR WATER SCIENTIST** with an image of a smiling water droplet wearing safety goggles.

***Teacher:** "Incredible job, team! You sorted all the objects and solved Chiku's courtyard mystery! Because of your amazing observation skills, Planrs Academy is proud to award you the official Junior Water Scientist Badge! Wear it with pride!"*

8.3 Extended Home Mission Assignment

***Teacher:** "But wait, your scientific journey isn't over yet! Here is your home mission: With the help of your mom or dad, explore your kitchen or balcony. Find one new object that floats and one that sinks. Draw them inside your Planrs Lab Book and show it to me in our next class! Keep testing, keep guessing, and always stay curious! See you next time! Bye-bye!"*

9. Minimum Quality Assurance & Production Standards

To ensure content remains premier, accessible, and high-impact across the Planrs Academy platform, all assets must meet these compliance criteria:

9.1 Media Production Technical Constraints

- **Video Output:** Minimum resolution must be 1920x1080p Full HD at 30 frames per second. Video color spaces should use vivid, high-saturation profiles to appeal to younger audiences.
- **Audio Mastering:** Voiceover tracks must be clear, center-panned, and free of background noise. The target master volume should be -12dB, with background instruments lowered to a soft -26dB padding level.
- **Interactive Code Performance:** UI elements must load within 1.5 seconds over standard 3G/4G connections. Touch targets must be at least 80x80 pixels to accommodate younger children's motor skills.

9.2 Technical Compliance Matrix

Compliance Vector	Standard Checkpoint Requirement	Verification Method
Jargon Control	Zero instances of advanced terms like density, buoyancy, or mass.	Automated text parsing review.
Material Safety	No glass, sharp items, or toxic chemicals in home activities.	Safety board review step.
Cultural Check	Examples use common Indian terms (e.g., Katori, Chammach, Aaloo).	Local design panel audit.
App Latency	Visual responses to taps or drags must trigger within 100ms.	System response testing.

FINAL AUDIT COMPLIANCE NOTE

Any content package that fails to meet these technical criteria or accidentally includes advanced jargon will be sent back for revision automatically. Adherence to these guidelines is mandatory for production authorization.